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Intent Integrity: CIA and Non-Repudiation as Properties of Intent, Not Data

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Abstract

The information-security canon names confidentiality, integrity, and availability (CIA), plus non-repudiation, as the load-bearing properties of **data**. Decades of operational practice — at the protocol level, at the storage level, at the wire level — operate against this property set. Governed agentic systems require an analogous property set applied to a different artifact: ***intent***, the operator-pinned envelope that the system is supposed to remain aligned with across its operating horizon. Intent-CIA-NR is not derivable from data-CIA-NR; the two property sets operate on different artifacts with different threat models, different verification primitives, and different operational disciplines.

This paper specifies intent-CIA-NR as the security model that the intent horizon — named in Paper One of this series — operationally requires. Confidentiality of intent is the property that **only the operator's authorized governance ceremony** can read or modify the pinned horizon. Integrity of intent is the property that the system's reasoning chain preserves the operator-pinned commitments end-to-end through every micro-decision the agentic execution layer makes. Availability of intent is the property that the operator can recover, in real time, the system's current intent — **not** its current output, **not** its current state, but the envelope the system is currently treating as authoritative. Non-repudiation of intent is the property that every horizon-event (pin, drift, storm, recalibration) is signed, logged, and cryptographically attributable to the governance authority that issued it.

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ABSTRACT

The information-security canon names confidentiality, integrity, and availability (CIA), plus non-repudiation, as the load-bearing properties of *data*. Decades of operational practice — at the protocol level, at the storage level, at the wire level — operate against this property set. Governed agentic systems require an analogous property set applied to a different artifact: **intent**, the operator-pinned envelope that the system is supposed to remain aligned with across its operating horizon.¹ Intent-CIA-NR is not derivable from data-CIA-NR; the two property sets operate on different artifacts with different threat models, different verification primitives, and different operational disciplines.

This paper specifies intent-CIA-NR as the security model that the intent horizon — named in Paper One of this series — operationally requires. Confidentiality of intent is the property that *only the operator's authorized governance ceremony* can read or modify the pinned horizon. Integrity of intent is the property that the system's reasoning chain preserves the operator-pinned commitments end-to-end through every micro-decision the agentic execution layer makes. Availability of intent is the property that the operator can recover, in real time, the system's current intent — *not* its current output, *not* its current state, but the envelope the system is currently treating as authoritative. Non-repudiation of intent is the property that every horizon-event (pin, drift, storm, recalibration) is signed, logged, and cryptographically attributable to the governance authority that issued it.

The argument proceeds in five movements. Section 2 documents the failures of treating intent-security as data-security with intent-shaped variable names — five operational patterns in which data-level CIA is satisfied while intent-level CIA is violated invisibly. Section 3 specifies intent-confidentiality and intent-integrity, with the threat models and verification primitives each requires. Section 4 specifies intent-availability and intent-non-repudiation, with operational examples drawn from the heterogeneous federation documented in MR-P9.² Section 5 places intent-CIA-NR within the HGC³AE² role ordering and shows how the prior architecture's runtime mechanisms acquire enforcement specificity once intent is the artifact under security. Section 6 hands off to IH-3 (Confident Misalignment as Horizon Failure), where the diagnostic consequences of intent-integrity loss are developed.

The central claim is that **intent without a security model is a soft target**: any system that can satisfy data-CIA while operating against a tampered, drifted, or unrecoverable intent envelope is operationally compromised by the standards governed agentic operation requires. Intent-CIA-NR closes this gap.

HOW TO CITE

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1 Justin H. Kuiper, CISSP, *The Intent Horizon: A Structural Primitive for Governed Agentic Operation* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-horizon/.

2 Justin H. Kuiper, CISSP, *Operational Plan: A Heterogeneous Federation as Proof of the Mission-Ready Architecture* (v0.1-seed, 2026), nonsequitur.tech/white-papers/operational-plan/.



1. INTRODUCTION

The Intent Horizon paper (IH-1) named the temporal and conceptual envelope of operator intent as an architectural primitive of governed agentic operation.³ Naming the horizon makes the diagnostic and enforcement instruments of the prior canon operationally precise. It also surfaces a security question the prior canon left implicit: what protects intent itself? Information-security practice has a well-developed answer for data — confidentiality, integrity, availability, plus non-repudiation. The answer for intent is structurally different, and the differences are operationally consequential.

Consider three commonplaces from data-security practice:

Data confidentiality prevents unauthorized read. The threat is information disclosure; the controls are encryption at rest, encryption in transit, and access control. None of these protect intent. An agentic system whose operating intent is in plain text inside its prompt context, or in plain text inside its persistence tier, may have perfect data-confidentiality while suffering complete intent-disclosure to anyone who can read the system's working memory.

Data integrity prevents unauthorized write. The threat is tampering; the controls are signed payloads, content hashes, and integrity-preserving protocols. None of these prevent the system's reasoning chain from *internally drifting* across its operating horizon. Intent integrity is violated by drift even when every byte of data is uncorrupted.

Data non-repudiation binds an action to an actor. The threat is repudiation; the controls are cryptographic signatures and authenticated audit logs. None of these capture *the operator's intent state* at the time an action was committed. An action that is non-repudiable at the data level may have been performed against an intent envelope the operator never pinned.

The gap is not that data-CIA-NR is unimportant — it is — but that data-CIA-NR is **insufficient** for governed agentic operation. The architectural commitments of HGC³AE²⁴ and the runtime enforcement of the Skipjack Protocol⁵ depend on intent being a verifiable, recoverable, attributable artifact. The intent horizon makes that artifact specifiable; this paper specifies the security model that protects it.

Section 2 catalogues five failure patterns in which data-CIA is satisfied while intent-CIA is violated. Section 3 develops intent-confidentiality and intent-integrity as the two read/write properties of the intent artifact, with threat models and verification primitives. Section 4 develops intent-availability and intent-non-repudiation as the two operator-facing properties — the operator's ability to recover and to attribute intent state across the system's operating lifetime. Section 5 reads the four properties against HGC³AE²'s role ordering and against Skipjack's enforcement mechanisms. Section 6 implications + handoff.

³ Justin H. Kuiper, CISSP, *The Intent Horizon: A Structural Primitive for Governed Agentic Operation* (v0.1-seed, 2026), nonsequitur.tech/white-papers/intent-horizon/.

⁴ Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

⁵ Justin H. Kuiper, CISSP, *Agile Scrum, Agenticity, and the Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



2. PROBLEM SET: DATA-CIA-NR SUFFICIENT, INTENT-CIA-NR VIOLATED

The five failure patterns below are all observable in operational agentic systems whose data-level security is properly engineered. Each is a case where the data-security canon's controls are satisfied at the byte level while the system's operating intent has been compromised at the architectural level. The patterns are taxonomic, not exhaustive; the unifying property is that intent and data are different artifacts and require different security models.

2.1 PLAINTEXT-INTENT IN ENCRYPTED-TRANSPORT

The most operationally common pattern. The system's operating intent is encoded as plain text inside a prompt, a system message, or a tool-call argument. The transport channel is TLS-encrypted; the persistence tier is at-rest-encrypted; the access-control matrix is audited and enforced. From a data-security audit, the system is exemplary. From an intent-security perspective, the system is exposing its full operating envelope to any process or persona that can read the prompt context. Intent-confidentiality is violated invisibly: there is no failed-decryption event, no rejected-access event, no audit-log entry — because no data-security control was violated.

2.2 DRIFT-WITHOUT-MODIFICATION

An agentic system executes a multi-step task. Each step's input and output is data-integrity-preserved (signed payloads, content-hashes, append-only logs). Across N steps, the system's *operating interpretation* of the operator's pinned intent drifts in ways the data-integrity controls do not detect. At step N, the system is acting against an intent envelope different from the operator's commit, but no byte of data has been tampered with. The drift is real; the data-integrity audit finds nothing; the intent-integrity audit (if it existed) would find the breach.

2.3 RECOVERY-OF-OUTPUT, LOSS-OF-INTENT

The system encounters a substrate failure and recovers via the operational-plan mechanisms.⁶ The recovery is successful by the metrics that data-recovery practice tracks: persistent state is restored, in-flight work is replayed, observability resumes. What is *not* recovered is the operator's intent envelope as it stood at the point of failure: the system resumes operating against a default intent, or against the most-recently-cached intent, or against an intent reconstructed by inference from the persistent state. The recovery is successful by data-availability metrics and incomplete by intent-availability metrics. Operators who do not have intent-availability instruments cannot tell the difference until a horizon-storm fires.

2.4 UNREPUDIATED ACTION, UNATTRIBUTED INTENT

Every action the system performs is signed and audit-logged. The action's authority chain back to a human operator is cryptographically traceable. Non-repudiation is intact at the action level. Inside the action's authority chain, however, the *intent state* at the time of commit is not captured: the audit log records *what was done* but not *against what envelope*. A subsequent dispute about whether the action was within the operator's intent at commit time cannot be

⁶ Justin H. Kuiper, CISSP, *Operational Plan: A Heterogeneous Federation as Proof of the Mission-Ready Architecture* (v0.1-seed, 2026), nonsequitur.tech/white-papers/operational-plan/.



resolved from the audit log alone, because the log captures data-level non-repudiation but not intent-level non-repudiation.

2.5 PRIVILEGED-CONTEXT INFLATION

Operators sometimes grant the system temporary access to a privileged context — confidential customer data, restricted tooling, elevated permissions — for a specific task. The data-access controls correctly gate the access at the appropriate boundaries. What the controls do not enforce is that the system's *operating intent* remain confined to the privileged-context's scope after the privilege is revoked. An intent that inflated to use the privileged context, and is not deflated when the context is removed, will drift into operating-as-if-still-privileged states. Data-CIA is intact; intent-CIA is violated by inflation.

2.6 THE UNIFIED PROBLEM STATEMENT

The five patterns share a structure. Each is a case where the data-level security controls operate against the wrong artifact: bytes rather than envelope, payload rather than commitment, action rather than authority state. The unified problem is that intent is not data: it is the architectural envelope within which data is interpreted, and securing the bytes that encode the envelope does not secure the envelope itself. Intent-CIA-NR is the property set that protects the envelope.

3. INTENT-CONFIDENTIALITY AND INTENT-INTEGRITY

3.1 INTENT-CONFIDENTIALITY

Property. Only the operator's authorized governance ceremony can read or modify the pinned intent.

Threat model. Two distinct threats. *Inference threat*: an adversarial process reads the system's prompt context, persistent state, or tool-call arguments and reconstructs the operator's pinned intent. *Influence threat*: an adversarial process modifies the system's working interpretation of intent without modifying the data that encodes it (e.g., via prompt-injection, via context-window pollution, via tool-call manipulation).

Verification primitive. Intent-confidentiality is verified by *operator-side replay*: the operator periodically queries the system's intent-rehearsal function⁷ and compares the returned intent against the operator's pinned commit. Discrepancies indicate either drift (legitimate, requires recalibration) or breach (a process has modified the operating intent outside governance ceremony). The verification primitive is read-side, not write-side: the operator cannot prevent every breach attempt; the operator can detect every breach by structural comparison.

Failure mode. Inferred intent in adversarial hands constitutes a complete-knowledge attack on the system's operating envelope. The adversary can predict the system's responses, plan adversarial inputs against the predicted decision points, and exfiltrate the system's operating logic from the outside. The HGC³AE² role ordering's *humans govern* commitment depends on this property holding.

⁷ Justin H. Kuiper, CISSP, *The Comprehension Instrument for Agentic Systems* (v0.1-seed, 2026), nonsequitur.tech/white-papers/comprehension-instrument/.



3.2 INTENT-INTEGRITY

Property. The system's reasoning chain preserves the operator-pinned commitments end-to-end through every micro-decision the agentic execution layer makes.

Threat model. Three distinct threats. *Drift threat*: cumulative micro-decisions move the operating intent away from the pinned commit (the §2.2 pattern). *Substitution threat*: a process substitutes a different intent for the operator's pinned one (prompt injection, tool-call manipulation). *Confusion threat*: the system's reasoning chain conflates two distinct intents (one pinned by the operator, one inherited from a prior session or default) and operates against a synthetic merger of the two.

Verification primitive. Intent-integrity is verified by *chain-of-custody attestation* across the reasoning steps. Each agentic execution step emits a structured trace that names which intent the step was committed against; the chain is verifiable by replay of the trace against the operator's pin. The chain-of-custody is to intent what hash chains are to data: a cryptographic structure that lets a verifier confirm end-to-end preservation without consulting the system's internal state during execution.

Failure mode. Drift, substitution, and confusion failures of intent-integrity produce competent-looking output that is structurally misaligned with operator commitment. The system is not malfunctioning by any traditional metric — it is producing coherent, plausible, actionable output. The output is against the wrong envelope. The diagnostic consequence is the failure mode that MR-P1 names as confident misalignment;⁸ IH-3 develops the horizon-based account of this failure.⁹

3.3 THE DISTINCTION BETWEEN INTENT-INTEGRITY AND DATA-INTEGRITY

A worked example illustrates the distinction. An operator pins the system to “draft a brief about Topic A, citing only sources from Domain X”. Data-integrity controls ensure the byte sequence of the brief is what the system actually wrote and that no third party tampered with it. Intent-integrity controls ensure that as the brief drafted, the system stayed within “Topic A” and used only “Domain X” sources. Data-integrity would catch a man-in-the-middle altering the brief; intent-integrity would catch the system citing a Domain Y source because the LLM's training distribution made it seem natural to do so. Both controls are necessary; neither subsumes the other.

4. INTENT-AVAILABILITY AND INTENT-NON-REPUDIATION

4.1 INTENT-AVAILABILITY

Property. The operator can recover, in real time, the system's current intent — the envelope the system is currently treating as authoritative.

⁸ Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

⁹ Forthcoming IH-3 *Confident Misalignment (Horizon-Framed)* (in concurrent drafting; this dispatch). Re-describes MR-P1 through horizon lens; H-21 cite-without-redefinition discipline applied.



Threat model. Two distinct threats. *Black-box threat*: the system's intent state is not introspectible by the operator; the operator can read the system's outputs but cannot read the envelope the outputs are interpreted against. *Recovery-loss threat*: a substrate failure causes the system to lose the operator-pinned intent state; the system recovers operationally (per the operational plan¹⁰) but the recovered state does not include the lost intent envelope.

Verification primitive. Intent-availability is verified by *operator-side query*: at any moment, the operator can request the system's current intent envelope and receive a structured, comparable artifact. The artifact is not the system's output; it is the envelope under which the output was produced. The structural form of intent — pinned operator commitments + horizon boundaries — makes the artifact comparable to the operator's prior pin and inspectable for drift.

Failure mode. A system whose intent is not operator-recoverable is a system the operator cannot govern in real time. The HGC³AE² human-in-the-loop commitment depends on the operator's ability to *read* the system's intent state, not just receive its outputs. Intent-availability is the property that makes that read possible.

4.2 INTENT-NON-REPUDIATION

Property. Every horizon-event — pin, drift, storm, recalibration — is signed, logged, and cryptographically attributable to the governance authority that issued it.

Threat model. Three distinct threats. *Authorship-dispute threat*: a post-incident audit cannot determine which operator pinned which intent at which time. *Substitution threat*: an unauthorized party modifies the intent state with no audit trail attributing the modification. *Replay threat*: a prior intent state is replayed at a later time without operator authorization, and the system operates against the replayed (stale) intent.

Verification primitive. Intent-non-repudiation is verified by an *append-only horizon log* signed by the governance authority issuing each event. The log structure matches the audit-log discipline of MR-P9's combined security-and-provenance layer,¹¹ but is dedicated to horizon events rather than data events. The log is independent of the system's operational logs (which capture what the system did); the horizon log captures what envelope the operator pinned, when, and under what authority.

Failure mode. Without intent-non-repudiation, governance loses the ability to settle disputes about whether an action was within authorized intent. The system may have acted competently; the operator may have intended differently; the gap is unresolvable without an authoritative log of intent state at the time of action. The discipline matters most at the boundary cases — incident retrospectives, regulatory audits, persona-handoff events — where the question is not what was done but what was *intended*.

4.3 THE ASYMMETRY WITH DATA-NON-REPUDIATION

Data-non-repudiation binds an action to an actor. Intent-non-repudiation binds an intent commit to a governance authority. The two are not interchangeable: a data-signed action is non-repudiable about *what was done by whom*; an intent-signed pin is non-repudiable about *what envelope was authorized by whom*. A complete operational log captures both. A log that

10 Justin H. Kuiper, CISSP, *Operational Plan: A Heterogeneous Federation as Proof of the Mission-Ready Architecture* (v0.1-seed, 2026), nonsequitur.tech/white-papers/operational-plan/.

11 Justin H. Kuiper, CISSP, *Security and Provenance for Self-Hosted Agentic Systems* (v0.1-seed, 2026), nonsequitur.tech/white-papers/security-provenance/. (MAO-P9, cited here for combined-layer audit-log discipline.)



captures only data-non-repudiation cannot answer intent questions; a log that captures only intent-non-repudiation cannot answer execution questions.

5. INTENT-CIA-NR WITHIN HGC³AE² ROLE ORDERING

The HGC³AE² framework's role ordering — humans govern; context is curated; AI executes; master stamps last¹² — applies to intent-security as it applies to every architectural commitment.

Humans govern intent-CIA-NR. The four properties are operator-authority-bound. Confidentiality is granted at governance time (who may read the pin); integrity is bounded by governance ceremony (which decisions are within-pin vs. require re-pin); availability is operator-facing (the operator's ability to read current intent is a governance property, not a developer convenience); non-repudiation is governance-attributable (intent events are signed against governance authority, not against execution actors).

Curated context maintains the verification surface. The verification primitives of §3 and §4 — operator-side replay, chain-of-custody attestation, operator-side query, append-only horizon log — require curated context to function. The replay needs a comparable representation of pinned intent; the chain-of-custody needs structured traces emitted by the execution layer; the query needs an introspectible state model; the log needs a horizon-event taxonomy. Each is a curation discipline maintained at Skipjack cadence.¹³

Agentic execution operates under intent-CIA-NR constraints. The execution layer emits chain-of-custody traces; respects intent-confidentiality boundaries (does not log pinned intent into observability surfaces an adversary may read); supports the introspection query (always able to report current operating intent on demand); cooperates with horizon-event logging (every recalibration is logged at the governance authority's signature).

Runtime verification reads intent-CIA-NR signals. Verification at runtime is the continuous reading of the four properties' status. A drop in any property indicates an architectural fault that fires Skipjack mechanisms: confidentiality breach fires structured-decomposition fail-closed; integrity drift fires iterative-feedback recalibration; availability loss fires human-in-the-loop escalation; non-repudiation gap fires context-integrity audit. The mapping between intent-CIA-NR properties and Skipjack mechanisms is sketched here and developed fully in IH-5.¹⁴

6. IMPLICATIONS AND CONCLUSION

Intent-CIA-NR closes a structural gap in governed agentic operation. The information-security canon has a security model for data; the architecture's primitive — operator intent within an operating horizon — requires its own security model. The four properties developed here (confidentiality, integrity, availability, non-repudiation, applied to intent) supply that model.

¹² Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

¹³ Justin H. Kuiper, CISSP, *Agile Scrum, Agentics, and the Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

¹⁴ Forthcoming IH-5 *Skipjack as Horizon Governance* (in concurrent drafting; this dispatch).



The implications for operational practice are concrete:

Audit instrumentation acquires a new dimension: alongside data-audit logs, governed agentic systems maintain a *horizon-audit log* recording every intent event with operator-authority signatures.

Adversarial threat models for agentic operation gain specificity: prompt injection is no longer “an input-validation problem” but an intent-integrity attack; context-window pollution is an intent-confidentiality attack; persona drift across handoffs is an intent-non-repudiation gap.

Recovery procedures change: post-failure recovery requires intent-recovery as a first-class step, not an inferential reconstruction from operational state.

Governance practice gains a verifiable artifact: the operator can demonstrate, in incident retrospective or regulatory audit, exactly what intent the system was operating against at the point of the event.

The central claim: **intent is the artifact governance protects; data is the artifact security protects; both property sets are required and neither subsumes the other**. Intent-CIA-NR makes the operator’s authority over the operating horizon enforceable at runtime and auditable retrospectively.

This paper closes Paper Two of the *Intent Horizon Theory* series. IH-3 (Confident Misalignment, horizon-framed) develops the diagnostic consequence of intent-integrity failure: a system operating outside its horizon while reporting horizon-integral status to its operator.¹⁵ IH-4 (Epistemic Weather) specifies the diagnostic instruments operators consult to read intent-CIA-NR state in real time.¹⁶ IH-5 (Skipjack as Horizon Governance) names the operational gates by which intent-CIA-NR is enforced.¹⁷

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— end v0.1-seed draft —

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¹⁵ Forthcoming IH-3 *Confident Misalignment (Horizon-Framed)* (in concurrent drafting; this dispatch). Re-describes MR-P1 through horizon lens; H-21 cite-without-redefinition discipline applied.

¹⁶ Forthcoming IH-4 *Epistemic Weather* (in concurrent drafting; this dispatch).

¹⁷ Forthcoming IH-5 *Skipjack as Horizon Governance* (in concurrent drafting; this dispatch).